

Engine Spec — Section Signup

CS 3540 · Game Programming · Fall 2026 · Professor Hunter · [uvucs3540/engine-spec](#)

One section each. First claim wins. S00–S03 are instructor-owned; these fifteen are yours. This sheet is a **hold, not the claim** — make it real with a PR adding your name to `spec/OWNERS.md`, **due Sun Sep 13**.

Read `spec/S00-overview.md` first, then the cheat sheet for the section you want. Pick the one **your own game needs**. The three marked **heavy** carry more work and pair well — take one with someone if you would rather. You are signing up for: prose independent builds agree on · conformance vectors · fixing it when builds diverge · a 15-min talk, weeks 9–16.

\$	Section	Read first	Name
S04	Transform hierarchy and scene graph	scene-graph-transforms	
S05	Entity and component store	entity-component-store	
S06	Event bus and messaging	game-programming-patterns	
S07	Input abstraction and mapping	game-loop-and-time	
S08	Asset handles, caching, provenance	asset-pipeline-and-provenance	
S09	Renderer seam and 2D backend	2d-rendering	
S10	3D backend (WebGL2) · heavy	3d-rendering-webgl · shaders-and-materials	
S11	Collision and spatial partition	collision-and-spatial-partition	
S12	Pathfinding, navigation, steering	pathfinding-and-navigation	
S13	Game AI: FSM, behavior trees, utility · heavy	game-ai-behavior	
S14	Procedural generation	procedural-generation	
S15	Narrative: dialogue graph and quest state	storytelling-in-games	
S16	Generator seam: LLM and asset providers	local-llm-in-games	
S17	Transport seam: Local and Pear · heavy	netcode · p2p-pear-holepunch	
S18	Audio and procedural music	audio-and-procedural-music	

S16 and S17 are where this course's two hardest constraints live — a language model inside a frame budget, and multiplayer with no server. More interesting than they look.